

NWS, Network Statistics, System Administrator Guide

System Administrator Guide

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1 About this document

1.1 Purpose

This document describes system administration tasks for Performance Management Solution, Network Statistics (NWS). NWS is an application suite containing the SMIA, SGw, SDM, PDM, PSA, NWS-AM, NWS-AP, NWS-AG, NWS-AW, NWS-AC, MIA and PMS applications. The purpose of this document is to provide the reader with complementary information when working with NWS support.

The Performance Management Solution, Network Statistics contains various applications with their individual System Administrator guides. This document addresses the configuration and troubleshooting of different applications at NWS level. The configuration and troubleshooting for each individual application can be found in SMIA, Statistical Measurement Initiation and Administration, System Administrator Guide, SGw, Statistical Gateway, System Administrator Guide, SDM, Statistical Data Mart, System Administrator Guide, PDM, Performance Data Mediation, System Administration Guide and PSA, Performance Statistical Alarms, System Administrator Guide and PM, Subscription Profiles and Performance Monitoring, System Administration Guide.

1.2 Target Group

The intended target group for this document is:

- System Administrator

1.3 Prerequisites

It is assumed that the user of this document:

- Is familiar with UNIX
- Is familiar with OSS
- Is familiar with Business Objects (BO)

1.4 Typographic Conventions

Table 1 *Typographic Conventions*

Convention	Description	Example
User Input	A command that you must enter in a Command Line Interface (CLI) exactly as written.	<code>cd \$HOME</code>
Command Variables	Command variables, the values of which you must supply.	<code><home_directory></code>
GUI Objects	GUI objects, such as menus, fields, and buttons.	On the File menu, click Exit .
Key Combinations	Key combinations.	Press Ctrl+X to delete the selected value. ⁽¹⁾
System Elements	Command and parameter names, program names, path names, URLs, and directory names	The files are located in <code>c:/HOME/bin</code> . ⁽²⁾
Output Information	Text displayed by the system	System awaiting input
Code Examples	Code examples	<code>stat char* months[] = \</code> <code>{ "Jan", "Feb" }⁽³⁾</code>

(1) The plus sign (+) indicates that you must press the keys simultaneously.

(2) The use of forward slash(/) is for Unix Systems, PC systems use the back slash(\)

(3) The back slash (\) is used to show where long lines are split.

2 Network Statistics, System Administration

NWS provides a uniform and common performance management solution to all three different SNMs, GSM RAN, CORE and WCDMA RAN.

NWS is a solution for statistical performance management in OSS. It consists of the following applications:

- Statistical Measurement Initiation and Administration (SMIA)
- Statistical Gateway (SGw)
- Statistical Data Mart (SDM)
- Performance Statistical Alarms (PSA)
- Performance Data Mediation (PDM)
- Network Statistics, Analyzer (NWS-AM, NWS-AP, NWS-AG, NWS-AW, NWS-AC)
- Measurement Initiation and Administration (MIA)
- Performance Management Subsystem (PMS)

SMIA provides the measurement initiation and administration support for STS (on the SP and AP) and OMS for AXE based nodes.

SGw collects the network statistical data from various NEs and formats the files to BCP and ASN.1 format

SDM is a data mart that is designed for the fast retrieval of performance data. SDM stores the data in the database for the users of OSS.

PSA generates performance statistical alarms for traffic related problems within a network, for example Sleeping Cell, Congestion, Dropped Call Rate and so on.

PDM provides measurement initiation and collection of statistical data from J20 GGSN and other IP nodes and formats files to BCP and XML format.

NWS-AM, NWS-AP, NWS-AG, NWS-AW, NWS-AC is a report generation module. It has predefined reports that support the operator in analyzing the behavior and performance of the network.

The NWS workflow from initiation of measurements to generation of reports for different Network Elements is described in the Figure 1 on page 4 and Table 2 on page 4.

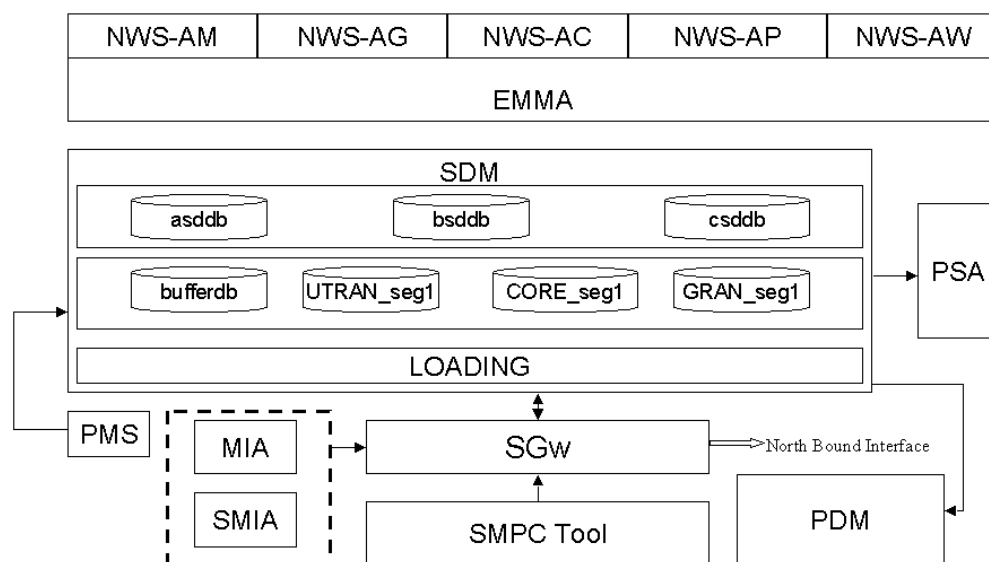


Figure 1 The NWS Architecture

Table 2 Applications Used in the Flow from Initiation to Generation of Reports for Different Network Elements

NE Type	Initiation of measurements	Collection of data	Parsing of data	Loading of data in databases	Consumers (of data in databases)	Reference
BSC/TRC (IOG and APG 40)	SMIA	SGw	SGw	SDM	NWS-AG, PSA	For IOG, refer Figure 2 on page 7. For APG40, refer Figure 3 on page 8.
MSC/VLR (IOG and APG 40)	SMIA	SGw	SGw	SDM	NWS-AG, NWS-AC, PSA	For IOG, refer Figure 2 on page 7. For APG40, refer Figure 3 on page 8.
HLR	SMIA	SGw	SGw	SDM	NWS-AG, PSA	

Table 2 Applications Used in the Flow from Initiation to Generation of Reports for Different Network Elements

NE Type	Initiation of measurements	Collection of data	Parsing of data	Loading of data in databases	Consumers (of data in databases)	Reference
SMPC	SMPCTool	SGw	SGw	SDM	NWS-AM, PSA	Refer Figure 4 on page 9.
WPP-GSN	MIA	SGw	SGw	SDM	NWS-AP, PSA	Refer Figure 5 on page 10
M-MGw	MIA	SGw	SGw	SDM	NWS-AC, PSA	Refer Figure 6 on page 11.
BGw/IPMM	MIA	SGw	SGw	SDM	NWS-AP, PSA	For BGw. refer Figure 7 on page 12
J20 GGSN and other IP Nodes	PDM	PDM	PDM	SDM	NWS-AP, PSA	Refer Figure 8 on page 13.
RNC, RBS and RXI	PMS	PMS	SDM	SDM	NWS-AW, PSA	Refer Figure 9 on page 14.

2.1 Directory Structure

For details regarding the directory Structures of individual applications, refer SMIA, Statistical Measurement Initiation and Administration, System Administrator Guide, SGw, Statistical Gateway, System Administrator Guide, SDM, Statistical Data Mart, System Administrator Guide , PDM, Performance Data Mediation, System Administration Guide and PSA, Performance Statistical Alarms, System Administrator Guide and PM, Subscription Profiles and Performance Monitoring, System Administration Guide.

2.2 Scripts and Program

For details regarding the Scripts and Programs pertaining to individual applications, refer SMIA, Statistical Measurement Initiation and Administration,

System Administrator Guide, SGw, Statistical Gateway, System Administrator Guide, SDM, Statistical Data Mart, System Administrator Guide , PDM, Performance Data Mediation, System Administration Guide, PSA, Performance Statistical Alarms, System Administrator Guide and PM, Subscription Profiles and Performance Monitoring, System Administration Guide.

2.3 Data Flow in NWS

2.3.1 IOG Node

The flow of information/data required for generation of reports in NWS for IOG based NEs is explained in the figure below.

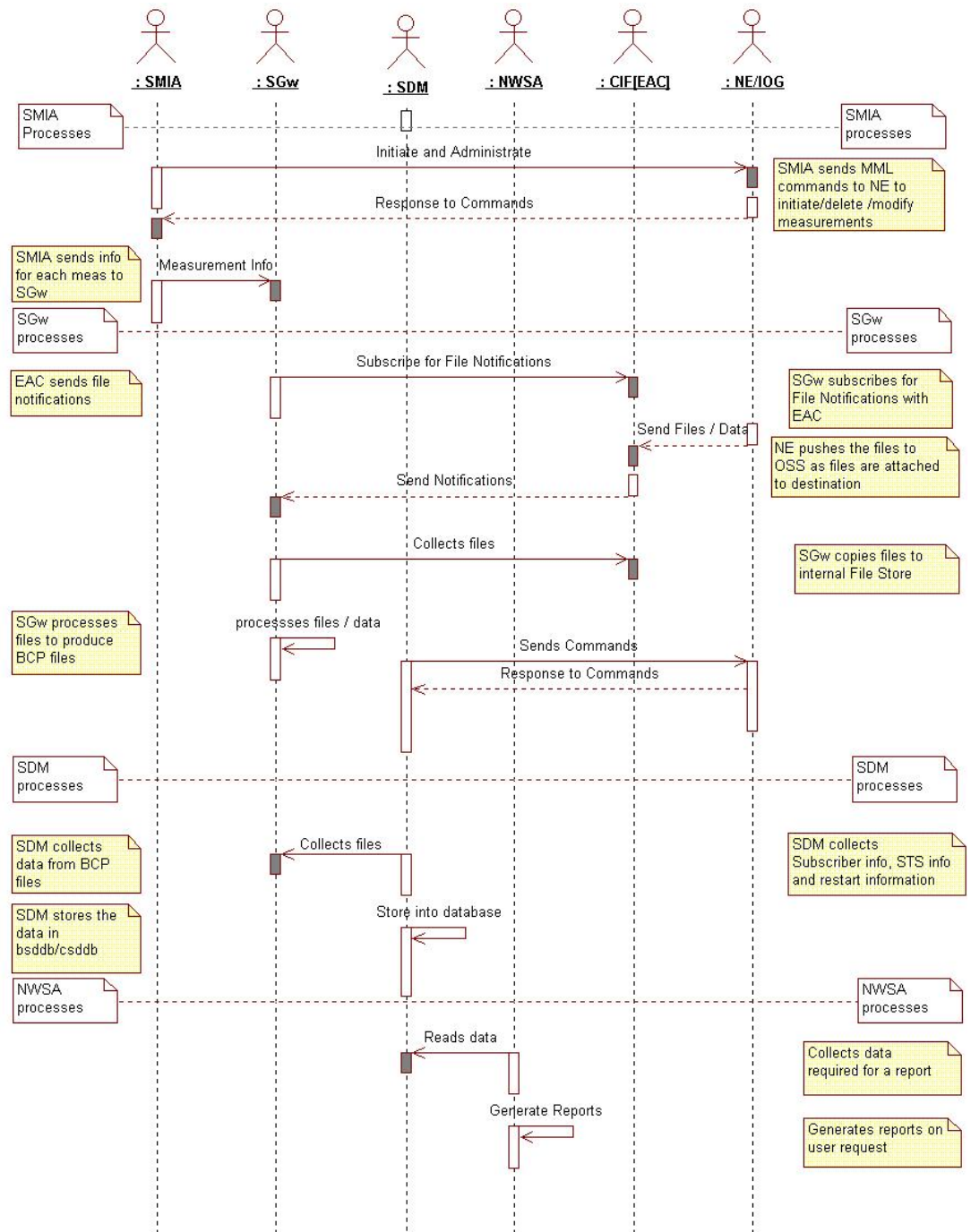


Figure 2 Data Flow for IOG based NEs

2.3.2

APG Node

The flow of information/data required for generation of reports in NWS for APG40 based NEs is explained in the following figure. SGW uses either FTP or SFTP for collection of files from APG 40.

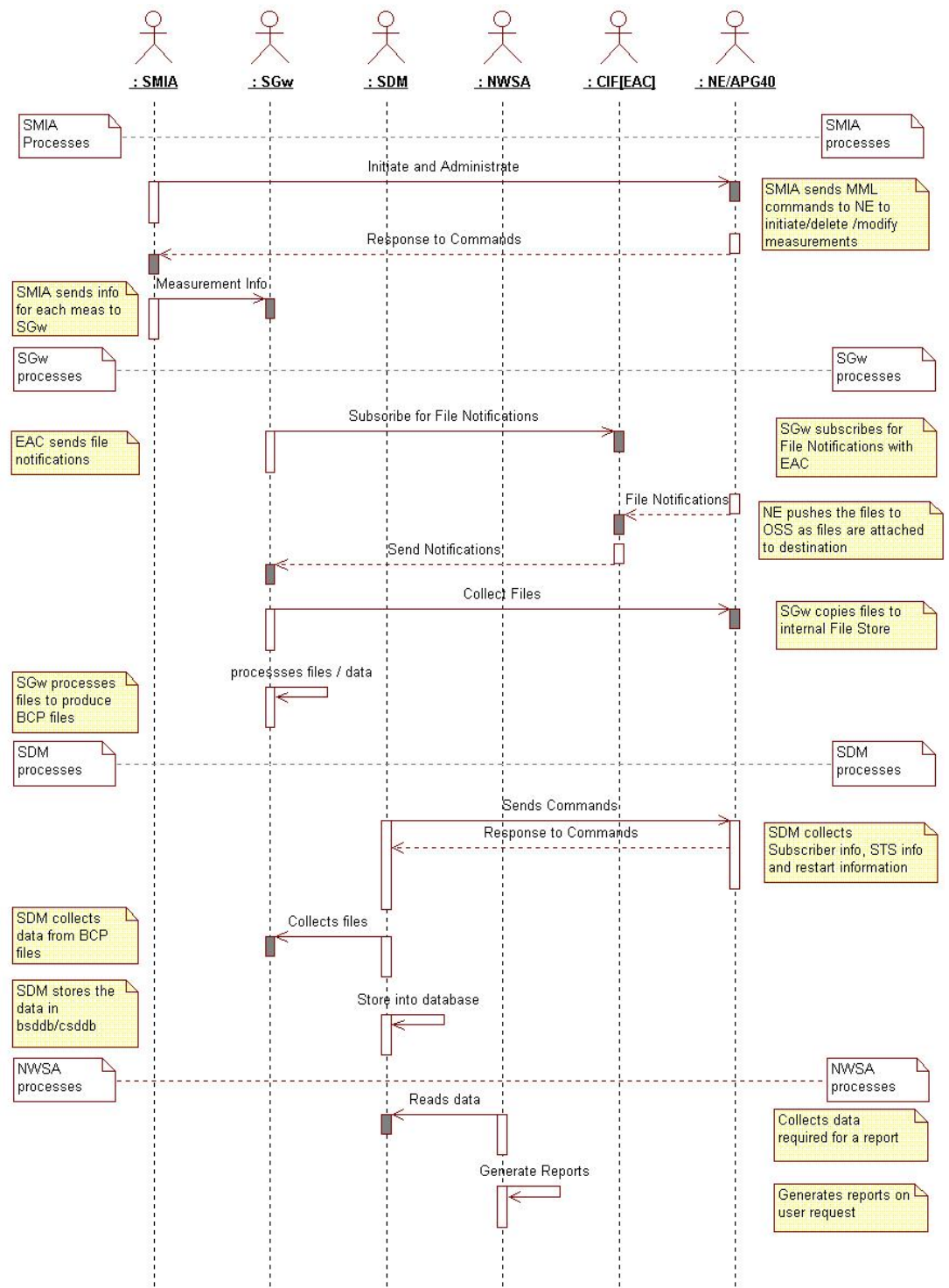


Figure 3 Data Flow for APG based NEs

2.3.3

SMPC Node

The flow of information/data required for generation of reports in NWS for SMPC Nodes is explained in the following figure.

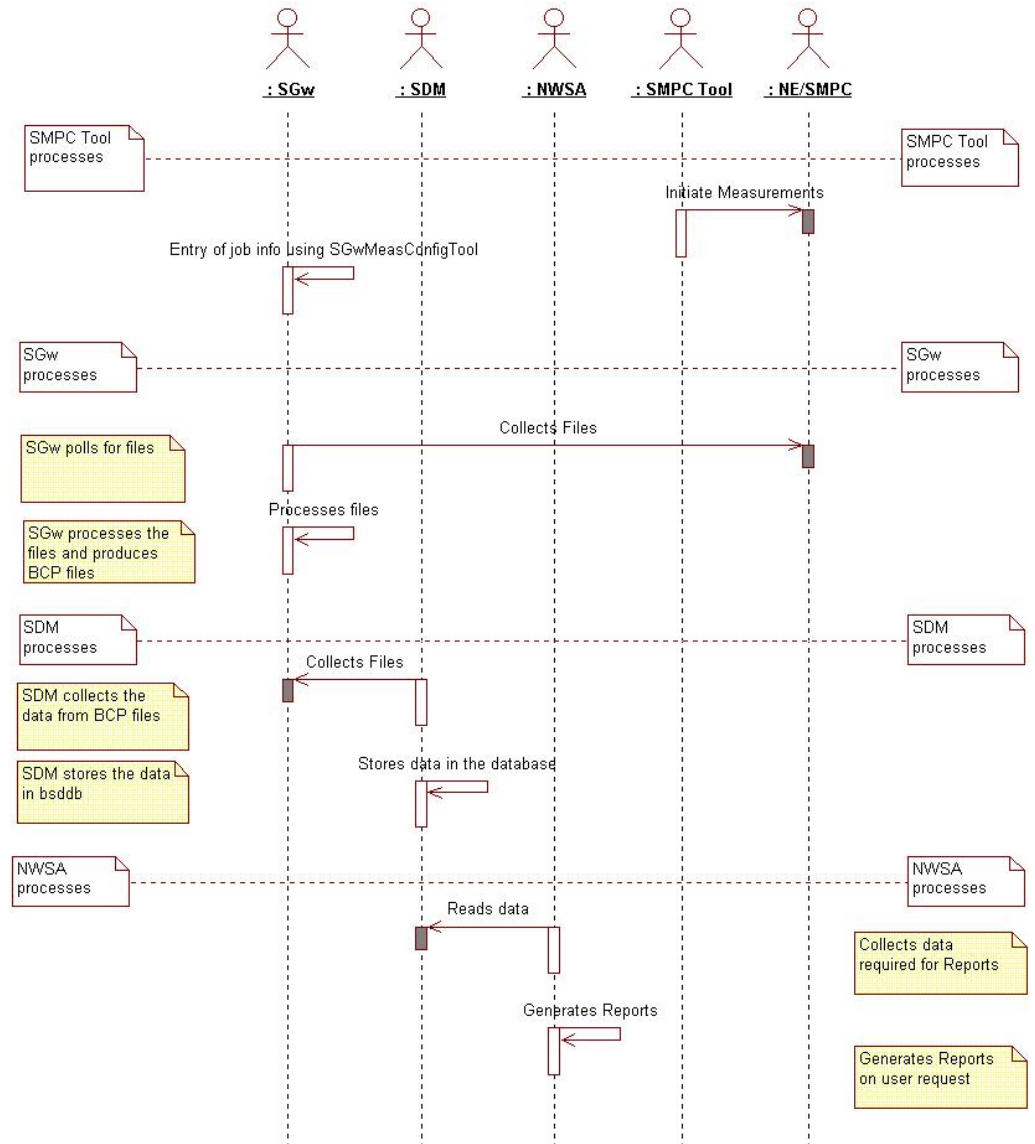


Figure 4 Data Flow for SMPC nodes

2.3.4

GSN Node

The flow of information/data required for generation of reports in NWS for GSN Nodes is explained in the following figure. SGw uses either FTP or SFTP for collection of files from GSN nodes.

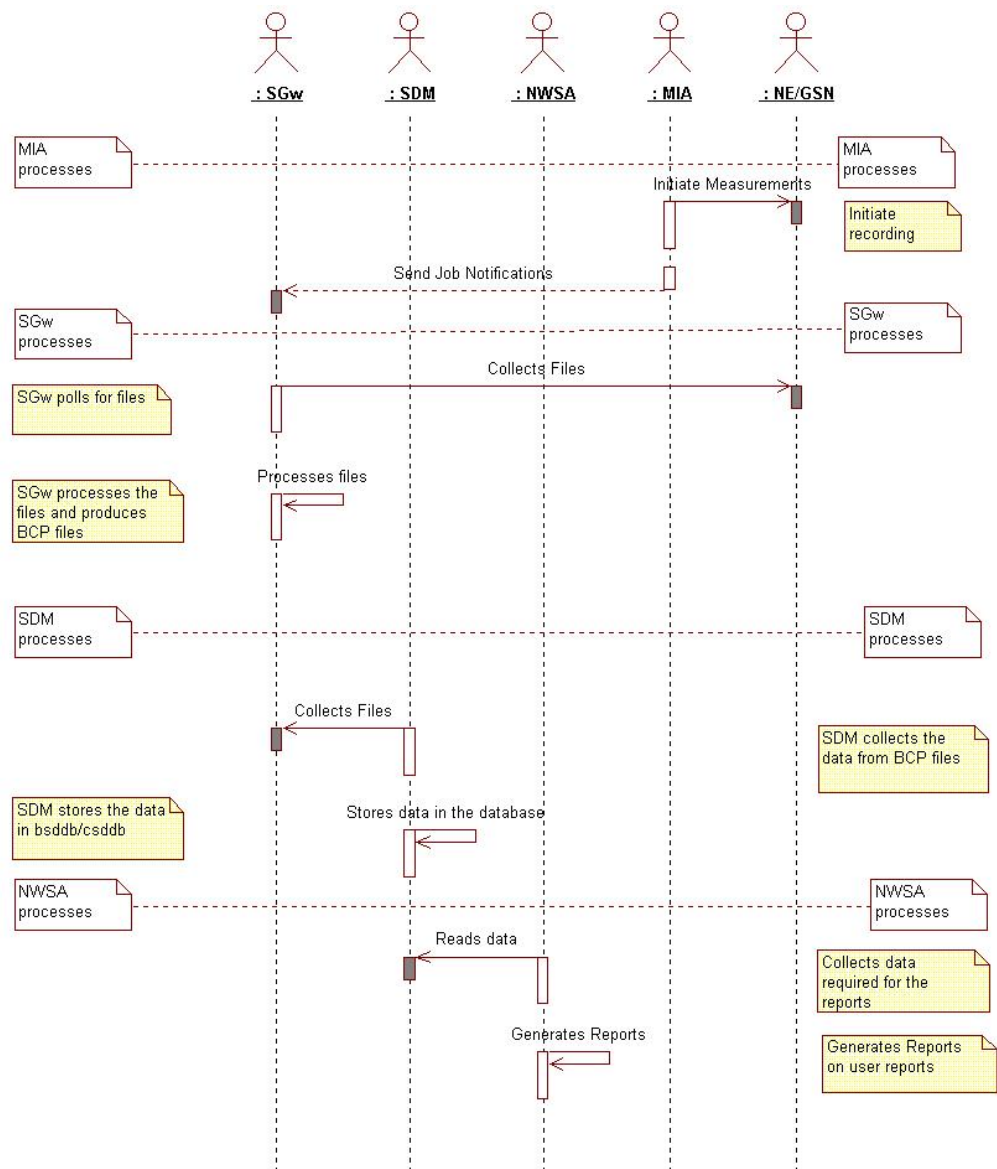


Figure 5 Data flow for GSN nodes

2.3.5 M-MGw Node

The flow of information/data required for generation of reports in NWS for M-MGw Nodes is explained in the following figure.

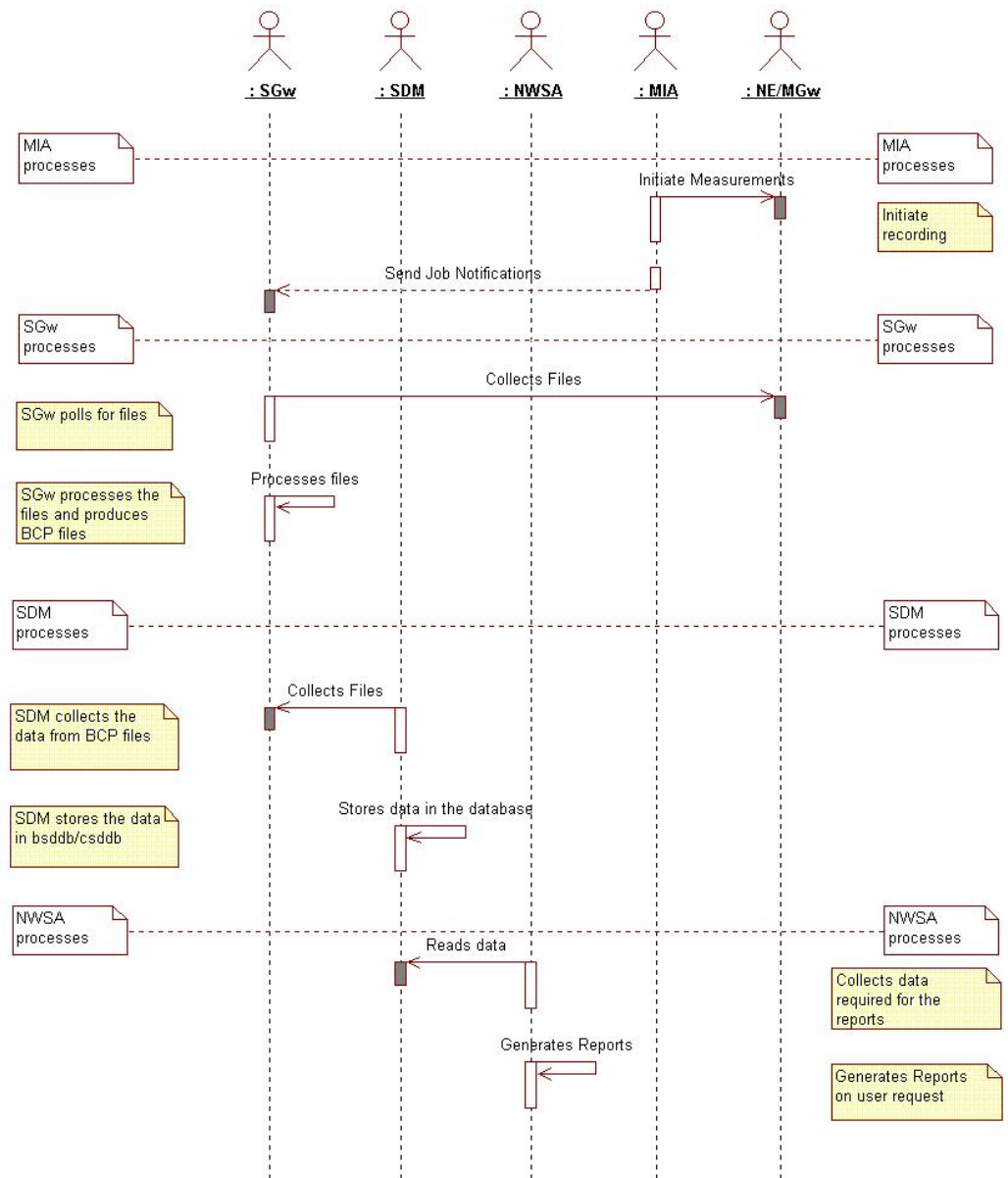


Figure 6 Data Flow for M-MGw nodes

2.3.6

BGw Nodes

The flow of information/data required for generation of reports in NWS for BGw nodes is explained in the following figure.

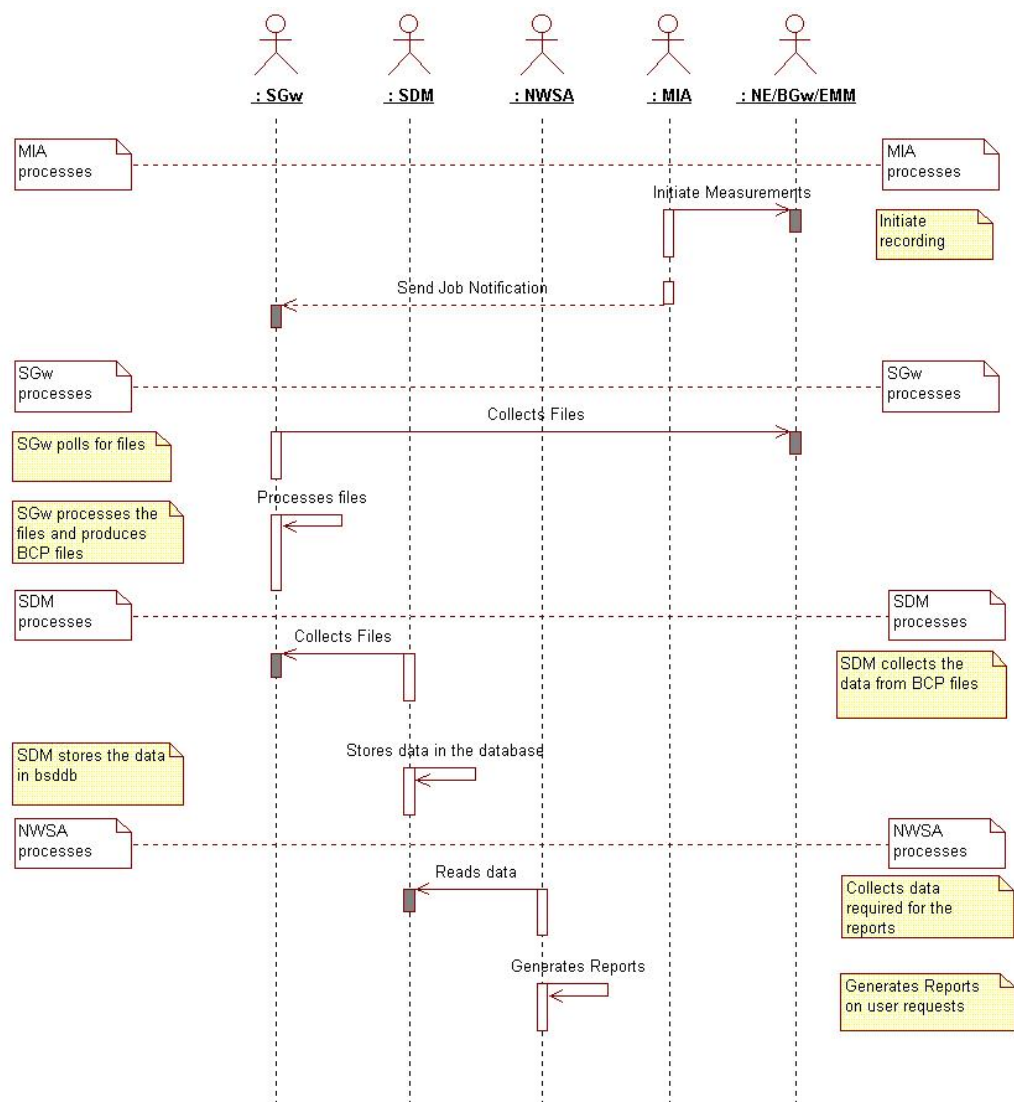


Figure 7 Data Flow for BGW nodes

2.3.7 J20 Node

The flow of information/data required for generation of reports in NWS for J20 Nodes is explained in the following figure.

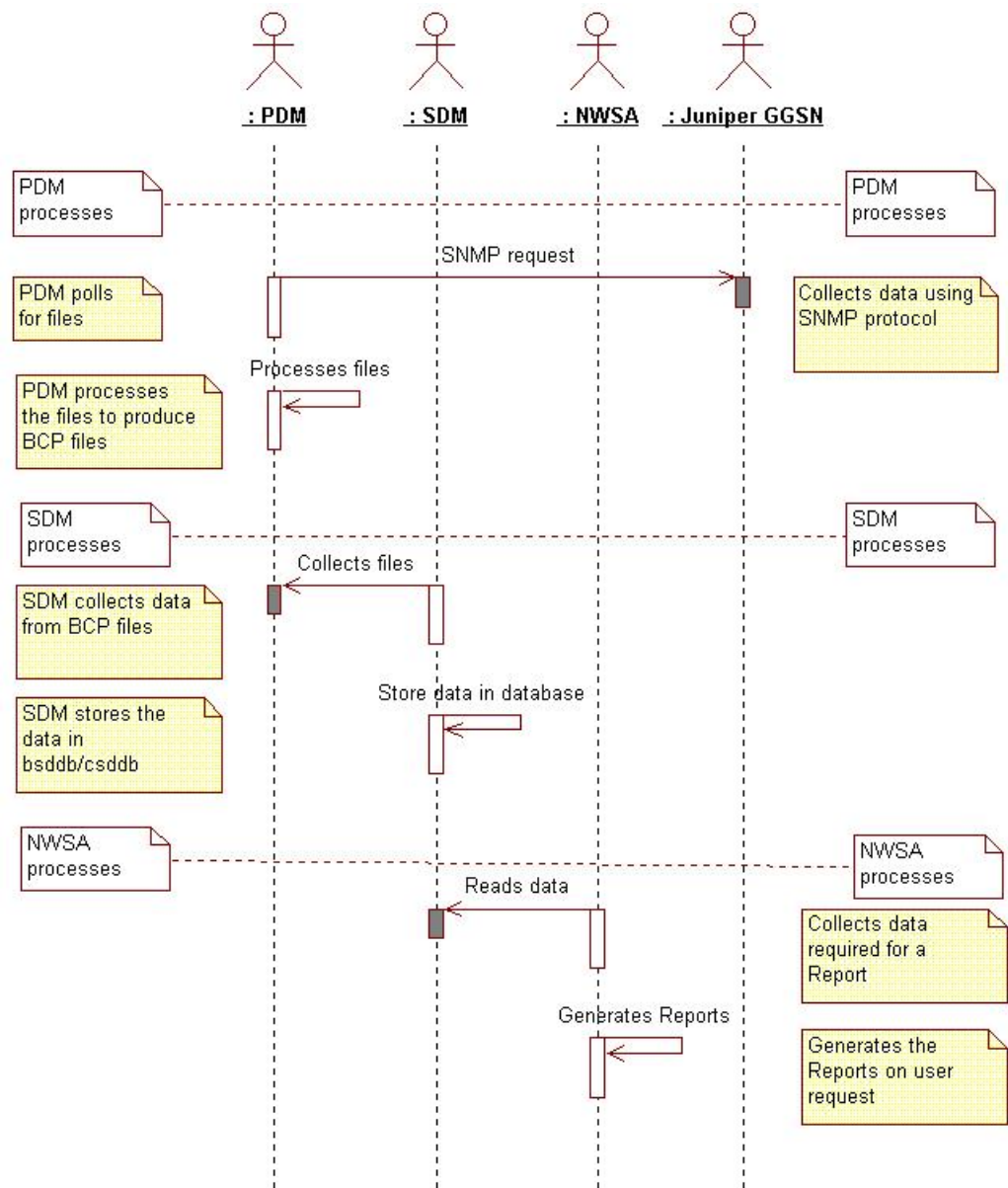


Figure 8 Data Flow for J20 based NEs

2.3.8 RBS/RNC/RXI Node

The flow of information/data required for generation of reports in NWS for RBS/RNC/RXI Nodes is explained in the following figure.

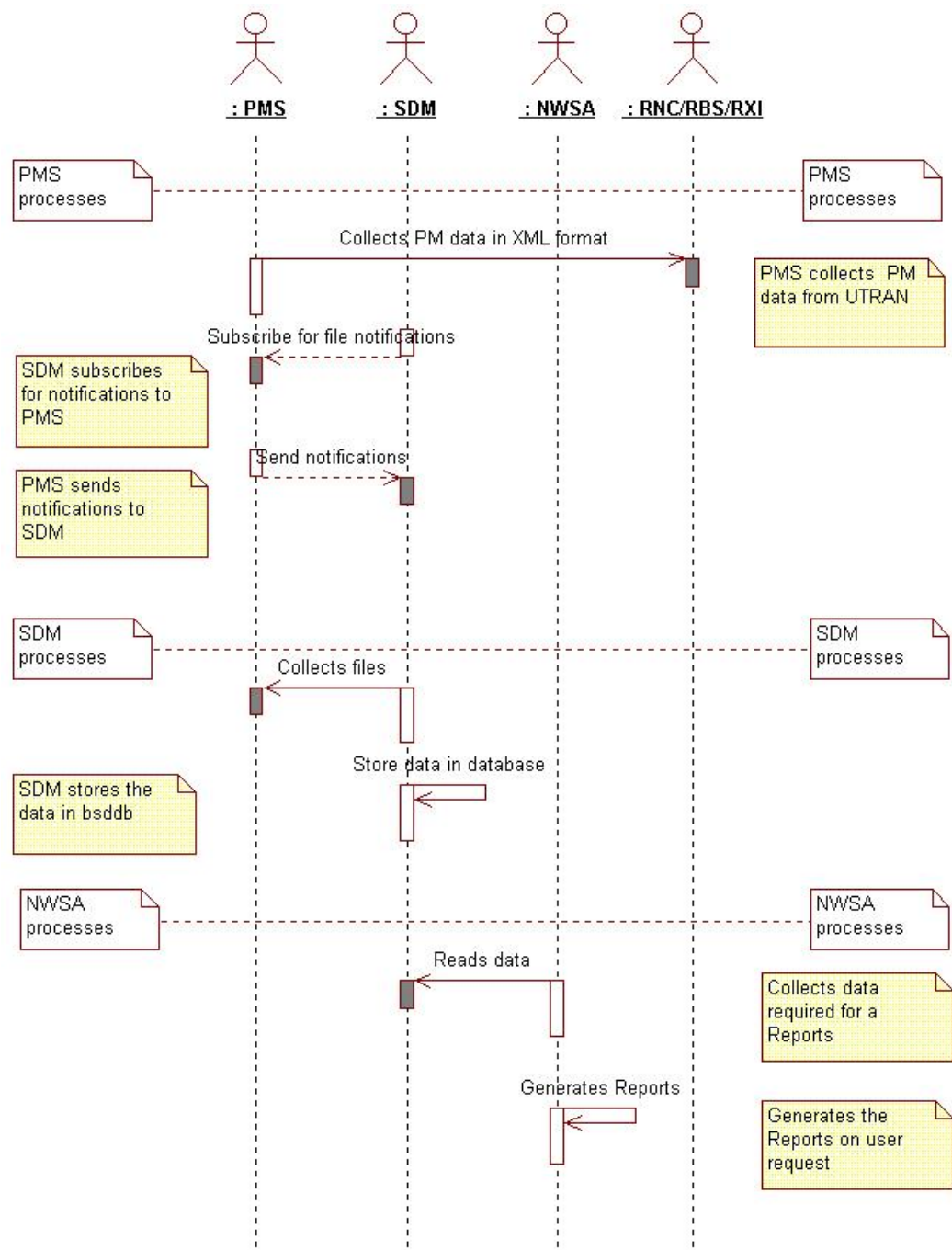


Figure 9 Data Flow for RBS/RNS/RXI nodes

3 Maintaining User Information

For details regarding the maintenance of user information, refer SMIA, Statistical Measurement Initiation and Administration, System Administrator Guide, SGW, Statistical Gateway, System Administrator Guide, SDM, Statistical Data Mart, System Administrator Guide, PDM, Performance Data Mediation, System Administration Guide, PSA, Performance Statistical Alarms, System Administrator Guide and PM, Subscription Profiles and Performance Monitoring, System Administration Guide.

4 Configuration Tasks

4.1 Environment Variables

For details regarding the Environment Variables used by the individual application, refer SMIA, Statistical Measurement Initiation and Administration, System Administrator Guide, SGw, Statistical Gateway, System Administrator Guide, SDM, Statistical Data Mart, System Administrator Guide, PDM, Performance Data Mediation, System Administration Guide, PSA, Performance Statistical Alarms, System Administrator Guide and PM, Subscription Profiles and Performance Monitoring, System Administration Guide.

4.2 PDB (Parameter Database) Maps

For details regarding the Parameter Database values used by individual application, refer SAG for individual application. For details on Parameter Maps used in SMIA, refer SMIA, Statistical Measurement Initiation and Administration, System Administrator Guide.

4.3 Processes in NWS

For details regarding the processes pertaining to individual applications refer SAG for individual application. Refer SMIA, Statistical Measurement Initiation and Administration, System Administrator Guide, SGw, Statistical Gateway, System Administrator Guide, SDM, Statistical Data Mart, System Administrator Guide, PDM, Performance Data Mediation, System Administration Guide, PSA, Performance Statistical Alarms, System Administrator Guide and PM, Subscription Profiles and Performance Monitoring, System Administration Guide.

4.4 Databases in NWS

For details regarding the databases pertaining to individual applications refer, individual application SAG. Refer SMIA, Statistical Measurement Initiation and Administration, System Administrator Guide, SDM, Statistical Data Mart, System Administrator Guide, PDM, Performance Data Mediation, System Administration Guide, PSA, Performance Statistical Alarms, System Administrator Guide and PM, Subscription Profiles and Performance Monitoring, System Administration Guide.

4.5 Error reports

For details regarding the Error Reports pertaining to individual applications, refer SMIA, Statistical Measurement Initiation and Administration, System Administrator Guide, SGw, Statistical Gateway, System Administrator Guide, SDM, Statistical Data Mart, System Administrator Guide , PDM, Performance Data Mediation, System Administration Guide, PSA, Performance Statistical Alarms, System Administrator Guide and PM, Subscription Profiles and Performance Monitoring, System Administration Guide.

5 Logging and Tracing

5.1 Logs

For details on logs pertaining to individual application, refer SMIA, Statistical Measurement Initiation and Administration, System Administrator Guide, SGw, Statistical Gateway, System Administrator Guide, SDM, Statistical Data Mart, System Administrator Guide , PDM, Performance Data Mediation, System Administration Guide, PSA, Performance Statistical Alarms, System Administrator Guide and PM, Subscription Profiles and Performance Monitoring, System Administration Guide.

5.2 Tracing

For details on trace pertaining to individual application, refer SMIA, Statistical Measurement Initiation and Administration, System Administrator Guide, SGw, Statistical Gateway, System Administrator Guide, SDM, Statistical Data Mart, System Administrator Guide , PDM, Performance Data Mediation, System Administration Guide, PSA, Performance Statistical Alarms, System Administrator Guide and PM, Subscription Profiles and Performance Monitoring, System Administration Guide.

6 Standard Maintenance Procedures

For details regarding the Standard Maintenance Procedures pertaining to individual applications, refer SMIA, Statistical Measurement Initiation and Administration, System Administrator Guide, SGw, Statistical Gateway, System Administrator Guide, SDM, Statistical Data Mart, System Administrator Guide, PDM, Performance Data Mediation, System Administration Guide, PSA, Performance Statistical Alarms, System Administrator Guide and PM, Subscription Profiles and Performance Monitoring, System Administration Guide.

7 Corrective Maintenance Procedures

For details regarding the Corrective Maintenance Procedures pertaining to individual applications, refer SMIA, Statistical Measurement Initiation and Administration, System Administrator Guide, SGw, Statistical Gateway, System Administrator Guide, SDM, Statistical Data Mart, System Administrator Guide , PDM, Performance Data Mediation, System Administration Guide , PSA, Performance Statistical Alarms, System Administrator Guide and PM, Subscription Profiles and Performance Monitoring, System Administration Guide.

8 Troubleshooting

8.1 Missing Files from UTRAN NEs

Performance Management Subsystem (PMS) collects raw data from the UTRAN NEs and notifies SDM about the arrival of data in XML-format.

Note: If notifications are not sent from PMS to SDM, then data will not be stored in SDM

Refer SDM, Statistical Data Mart, User Guide to check that notifications are working.

8.2 Missing data from AXE Network Element(s)

If data for a particular NE is missing in NWS, verify that the statistics are still collected in the AXE. When doing certain maintenance work in the AXE, for example a Function Change, the statistic collection will be stopped and needs to be restarted.

A Function Change on the AXE might also remove Object Types from the STS database. An STS measurement on an Object Type that has been removed will result in zero records for that Object Type.

If size of the measurement files (IOG STS/OMS and APG OMS) are not a multiple of 2048 bytes, there is probably something wrong with the measurement or the Object Types on the AXE from which the file was received.

8.2.1 Verify STS for IOG

- 1 Check if all the required Object Types are included in the STS database according to OSS-RC R3, AXE-IOG/APG Setup and also check the status of the recording.

IMLIT:SPG=0;

SDDOP:AREA=OPER,OBJTYPE=ALL;

- 2 Check the reports and files

SDFOP:RPTID=ALL,REFC;

A list containing file, reportid and Object Types per reportid is presented. FAULT CODE 36 means that no reportid is defined, nothing is defined to be measured.

SDTPP:MP=ALL;

END;

This command prints the list of STS Measurements. One of the printed parameter is the reportid. It is the same reportid from the previous command.

Use the printout of these commands to verify:

- whether the reportid(s) contain all required Object Types according to SDM, Statistical Data Mart, System Administrator Guide.
- the name of the STFIOP file that is used for every measurement.

3 Check the name of the transferred file

Normally the name of the file that is transferred to the OSS is the name found in the result of the SDFOP command, but sometimes the operator renames the file while transferring it. The real name is important, because that name must be inserted into the file `SGwMeasInfo.dat`. The `FILEID1` of the following command is the name of a subfile when transferring it via data link. If the `FILEID1` is empty, the file is not renamed during the transfer.

INFUP:FILE=<file from the SDFOP command above>;

4 Check the status of the file transfer and size of the file

Now we can check whether all subfiles have been transferred. In addition, the size of the files in blocks is displayed. If the size is 1 block, there is a problem:

- Object Types in the file are not included in the STS database
- STS Recording is OFF

INFSP:FILE=<transferred file name>,DEST=<dest>;

If the `<dest>` is unknown, use ILDNP MML command.

8.2.2 Verify STS for APG

Connect to APG node.

- 1 Check if all the required Object Types are included in the STS database according to OSS-RC R3, AXE-IOG/APG Setup and also the recording status.

aploc;

stmotls -l

exit;

The printout should contain: RECORDING ON and INCL YES for all required Object Types.

2 Check reports and report names

```
aploc;

stmrp -l -L

exit;
```

This command lists ALL reports in a long format. The list contains ReportId, ReportName and ObjectTypes per ReportId. Note the ReportId for the given ReportName.

3 Check the MPs

```
aploc;

stmmmp -l -L

exit;
```

This command prints the list of ALL MPs in long format. The list contains MPId, BeginTime, EndTime, Interval, ReportId, Status. Use the printout to verify whether or not the ReportIds obtained in the previous step are present here.

4 Verify the destination

```
aploc;

cdhls -l

exit;
```

This command lists all destinations defined. Check that the destination has following attributes.

```
-t          ftpv2
-c          r
-f          <OSS IP Address>
-x          50010
```

Verify the destination set. This command lists all the destination sets along with the destinations attached to it.

```
aploc;

cdhds1s -l
```

```
exit;
```

Verify whether or not the selected destination set is in the list.

5 Check TransferQueue and Destination

```
aploc;
```

```
afpls -l
```

```
exit;
```

This command lists all the TransferQueues and the attached Destinations. Check the TransferQueue is attached to the proper OSS Destination.

6 Check the files arrival

Go to the directory.

```
K:\AES\data\cdh\ftp\<Destination Name>\Ready\
```

Check whether or not directories are created when files arrive. The directory will be in the following format:

```
<TransferQueueName>-MP<mp Number>_<startDate>_<sequenceNum>
```

Example : A Measurement with Transfer Queue NWSSTS289324, with MP Number 1007 and start date & time 2002-08-12 13:15 will have directories like:

```
- NWSSTS289324-MP1007_200208121315_1
- NWSSTS289324-MP1007_200208121315_2
- NWSSTS289324-MP1007_200208121315_3
```

Go to these directories and check whether the files have arrived or not in the respective directories. The measurement file name starts with "C". If the file is missing check the status file present in the same directory.

Note: If SGW is collecting those files properly, those directories will be moved to:

```
K:\AES\data\cdh\ftp\<Destination Name>\Delete\
```

8.2.3 Verify OMS for IOG

Verify that the RELEASE parameter of TRARFILE, TRARTFILE and TRDIPFILE is set to YES


```
IOIFP:FILE=TRARFILE;
```

```
IOIFP:FILE=TRARTFILE;
```

```
IOIFP:FILE=TRDIPFILE;
```

RELEASE=NO may result in subfiles containing more than one measurement period. SDM does not support that.

8.2.3.1 Traffic on Routes

Investigate the result of **TRGRP:TRG=ALL;** and **TRTSP:MP=ALL;**

The first command prints the recording object type, scanning interval and recording objects for all recording groups. The second command prints the time schedules of the specified measuring programs.

Check that TRARFILE subfile is transferred:

```
INFSP:FILE=TRARFILE,DEST=<dest>;
```

If the <dest> is not known, use ILDNP MML command, which lists all the destinations attached to the NE.

8.2.3.2 Traffic Types

Investigate the result of **TRATP:MP=ALL;** If Traffic Measurement On Traffic Types is active, the result should be one measurement with IO=FILE.

FAULT CODE 13 means that no measurements are defined for Traffic Types. Check that the TRARTFILE is transferred:

```
INFSP:FILE=TRARTFILE,DEST=<dest>;
```

If the <dest> is not known, use ILDNP MML command, which lists all the destinations attached to the NE.

8.2.3.3 Traffic Dispersion

Investigate the result of **TRDGP:RG=ALL;** and **TRDMP:MP=ALL;**. For detailed information per MP:

```
TRDMP:MP=<MP Number>,data;
```

Check that the TRDIPFILE is transferred:

```
INFSP:FILE=TRDIPFILE,DEST=<dest>;
```

If the <dest> is not known, use ILDNP MML command, which lists all the destinations attached to the NE.

8.2.4 Verify OMS for APG

1 Verify definition of the files TRARFILE, TRARTFILE, TRDIPFILE

Verify if the definition of OMS files TRARFILE, TRARTFILE, TRDIPFILE in the CP is according to OSS-RC R3, AXE-IOG/APG Setup

```
aploc;

cpfls -l

exit;
```

This command lists all the Files. Check whether or not OMS files exist in this list.

2 Verify OMS files and destinations attached

```
aploc;

afpls -l <filename>

exit;
```

where filename is TRARFILE, TRARTFILE or TRDIPFILE.

This command lists the file details and destinations attached to the file. Check whether or not a proper OSS DESTINATION is attached to the file.

8.3 Missing data from non-AXE NEs

Verify the following for M-MGw, IPMM and SMPC nodes:

- Check if the scanners/ measurements are created on the node.
- Check if the files are being generated with appropriate contents and file naming conventions as per the file formats.
- Check if the UTC offset configuration on the node is synchronized with that Configuration Service.
- Check the connectivity towards the nodes and the user information in TSS.

Glossary

The OSS-RC Glossary is available in
Operations Support System (OSS) Glossary

Reference List

- [1] *Operations Support System (OSS) Glossary*, 0033-AOM 901 017/2
- [2] *SMIA, Statistical Measurement Initiation and Administration, System Administrator Guide*, 1543-APR 901 220
- [3] *SGw, Statistical Gateway, System Administrator Guide*, 1543-APR 901 432
- [4] *SDM, Statistical Data Mart, System Administrator Guide*, 1543-APR 901 011
- [5] *SDM, Statistical Data Mart, User Guide*, 1553-APR 901 011
- [6] *PSA, Performance Statistical Alarms, System Administrator Guide* , 1543-APR 901 175
- [7] *OSS-RC R3, AXE-IOG/APG Setup*, 1/1531-APR 901 122/2
- [8] *PDM, Performance Data Mediation, System Administration Guide*, 1543-APR 901 828
- [9] *PM, Subscription Profiles and Performance Monitoring, System Administration Guide*, 1/1543-APR 901 91/3